

TOWARDS GLOBAL GEOMETRY PRESERVATION IN TEXT EMBEDDING DISTILLATION

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ABSTRACT

Text embedding models derive much of their utility from the geometry they induce over a corpus. Distilling such a model should therefore align the student’s global geometry with the teacher’s. However, existing relational knowledge distillation (KD) typically constructs relations only among examples that share a mini-batch. The resulting supervision is selected by batch co-occurrence rather than teacher geometry, leaving most teacher-defined neighborhoods unobserved. We introduce **RIPPLE**, a corpus-anchored approach that decouples relation selection from mini-batch optimization: a sparse teacher graph determines which corpus relations to transfer, while each update evaluates only a compact set of texts. **RIPPLE** diffuses the graph to represent multi-hop structure, samples candidates from the resulting target mass, and supervises overlapping rows to couple independently normalized anchors. It requires neither online teacher inference nor task labels and adds no inference-time cost. Across nine datasets and three teacher–student pairs, **RIPPLE** achieves the best mean score among the evaluated compact students in all three settings, with the strongest gains on semantic textual similarity.

1 INTRODUCTION

Text embedding models map examples into a shared space in which semantic similarity can be recovered by simple geometric operations. Their usefulness therefore depends not only on individual embeddings, but also on how examples are arranged across a corpus: semantic neighbors should remain close, unrelated regions should remain separated, and local neighborhoods should connect in a meaningful way. Prior work has shown that pairwise similarities, relative geometry, and global topological summaries can all provide effective distillation targets (Tung & Mori, 2019; Park et al., 2019; Passalis et al., 2021; Kim et al., 2023; 2024). This makes *global geometry alignment* a natural objective for embedding distillation: a compact student should preserve the teacher’s corpus-level organization while reducing deployment cost.

For embedding models, relational KD is particularly appealing because it matches teacher and student similarities (Park et al., 2019; Passalis et al., 2021). In practice, however, these similarities are usually normalized over examples that happen to share a mini-batch. This creates a support mismatch. A *global teacher neighbor* is selected from the full corpus by teacher similarity, whereas a *batch-local comparison* is selected by the data loader. The objective aims to transfer corpus geometry, but observes relations through stochastic batch co-occurrence.

This mismatch is severe even under repeated reshuffling. Proposition 1 shows that an anchor observes only an $O(TB/N)$ fraction of its teacher neighbors after T epochs with batch size B . Under our training schedule, the expected coverage is 1.03%: approximately 99% of the teacher neighborhood never enters the realized relational objective. Figure 1 illustrates this mini-batch relation gap and measures the accompanying topology distortion.

We introduce **RIPPLE** to decouple *which relations provide supervision* from *how they are computed*. A cached teacher graph fixes corpus-level support; diffusion turns immediate neighborhoods into multi-scale profiles; and mass-aware sampling evaluates those profiles on compact candidate sets. Overlapping-row supervision then couples relations across anchors. Each update encodes only a small pool of texts, but its contents are selected by teacher geometry rather than batch co-occurrence.

Global geometry alignment $\longrightarrow$ **mini-batch relation gap** $\longrightarrow$ **teacher-anchored supervision**

Expected teacher-neighbor coverage: 1.03% observed; 98.97% unseen.

Figure 1: **Relational KD uses a local view of a global target.** Teacher neighborhoods are defined over the corpus, whereas conventional relational losses obtain their support from mini-batch co-occurrence. Proposition 1 quantifies the missing support; the motivating experiment measures its effect on teacher–student topology alignment.

Across nine datasets and three teacher–student pairs, RIPPLE achieves the best mean compact–student score in all three settings. Controlled comparisons separate the effects of corpus anchoring, diffusion, candidate sampling, and row supervision. Our contributions are:

- We identify and quantify a mini-batch relation gap that leaves most teacher-neighbor scores outside the realized relational objective.
- We introduce a corpus-anchored distillation method that combines teacher-graph diffusion, mass-aware candidate sampling, and cross-anchor row supervision.
- We connect the resulting objective to diffusion-geometry preservation and evaluate it across nine datasets and three teacher–student pairs.

2 RELATED WORK

Text embedding distillation. Knowledge distillation transfers the behavior of a large teacher to a compact student (Hinton et al., 2015). Language-model compression commonly matches output distributions, hidden states, or attention maps (Sanh et al., 2019; Jiao et al., 2020; Wang et al., 2020; Xu et al., 2024). Sentence- and retrieval-embedding methods instead distill the quantities used for similarity search. DistilCSE variants combine teacher supervision with contrastive objectives (Gao et al., 2021; Xu et al., 2023); EmbedDistill transfers relative query–document geometry and augments sparse regions of the data manifold (Kim et al., 2023); and recent systems explore embedding-, layer-, or token-level transfer across heterogeneous encoders (Lee et al., 2024; Zhang et al., 2024; Vujanic & Ruecksties, 2025; Truong et al., 2025; Dao et al., 2026; Akram et al., 2026). These methods establish strong embedding-specific transfer mechanisms, but do not treat anchor-conditioned corpus topology as the primary supervision object.

Geometry as distilled knowledge. A broad line of work supports the premise that a teacher’s organization of examples is itself transferable knowledge. Similarity-preserving KD matches pairwise activation similarities (Tung & Mori, 2019); RKD matches inter-example distances and angles (Park et al., 2019); and PKT aligns conditional affinity distributions (Passalis et al., 2021). Instance Relationship Graph distillation represents features as vertices and inter-instance relations as edges (Liu et al., 2019), while BINGO forms teacher-guided bags of related instances (Xu et al., 2022). Most directly, TopKD defines global topology through persistence diagrams that summarize the shape, connectivity, and multi-scale structure of a latent point cloud (Kim et al., 2024). Together, these results motivate geometry preservation as a distillation goal. RIPPLE specializes this goal to text encoders by defining geometry through anchor-wise teacher neighborhoods and their diffusion profiles over a fixed corpus.

The mini-batch relation gap. The existence of a relational loss does not ensure that it observes the relations that define corpus geometry. Similarity-preserving KD, for example, constructs its pairwise matrix from the current mini-batch (Tung & Mori, 2019), and NRKD searches for neighbors within the mini-batch similarity matrix (Gou et al., 2025). Teacher-guided grouping enlarges the relational context (Xu et al., 2022), whereas a global topological summary does not specify which corpus

neighbors supervise each anchor (Kim et al., 2024). We study this unresolved support question by making relational supervision anchor-specific and corpus-defined.

Graph geometry and diffusion. Neighborhood graphs have long been used to recover manifold structure, from Laplacian Eigenmaps to Diffusion Maps (Belkin & Niyogi, 2003; Coifman & Lafon, 2006). Heat-based constructions further provide multi-scale descriptions of geodesic geometry (Sun et al., 2009; Huguet et al., 2023). In graph learning, geometric KD aligns neural heat kernels between teacher and student graphs (Yang et al., 2022), and other graph-distillation methods transfer observed graph structure to graph models or graphless students (Tian et al., 2025; Zhang et al., 2022). RIPPLE differs in both setting and deployment: its graph is induced from cached text embeddings solely to define supervision, while the student remains a standard text encoder that requires neither corpus graph nor teacher at inference.

3 GLOBAL GEOMETRY AND THE MINI-BATCH RELATION GAP

Global geometry alignment. Let $\mathcal{D} = \{x_i\}_{i=1}^N$ be an unlabeled corpus. The teacher produces unit-normalized embeddings t_i and cosine scores $c_{ij}^T = \cos(t_i, t_j)$; the student analogously produces s_i and c_{ij}^S . We use *global geometry* to refer to teacher-defined relations over $\mathcal{D}$, including the immediate teacher neighborhood

$$\mathcal{N}_k^T(i) = \text{TopK}_{j \in \mathcal{D} \setminus \{i\}} c_{ij}^T \quad (1)$$

and the multi-hop organization of these neighborhoods. The goal is to align this corpus-level structure between teacher and student.

Batch-local relational KD. For a mini-batch $\mathcal{B} \ni i$, let $\mathcal{L}_{\mathcal{B}}(i) = \mathcal{B} \setminus \{i\}$ denote the available comparison support. A representative relational objective compares

$$p_i^{T,\mathcal{B}}(j) \propto e^{c_{ij}^T/\tau_0}, \quad p_i^{S,\mathcal{B}}(j) \propto e^{c_{ij}^S/\tau_0}, \quad j \in \mathcal{L}_{\mathcal{B}}(i), \quad (2)$$

through $D_{\text{KL}}(p_i^{T,\mathcal{B}} \| p_i^{S,\mathcal{B}})$. Although the target is global geometry, the support of this loss is determined by the data loader and is independent of $\mathcal{N}_k^T(i)$.

Proposition 1 (Batch-local support blindness). *Assume that each epoch partitions the corpus uniformly into batches of size B . For an anchor i , let*

$$\mathcal{O}_i^{(T)} = \bigcup_{e=1}^T \mathcal{L}_{\mathcal{B}_e}(i) \quad (3)$$

be the examples compared with i during T independently reshuffled epochs. Then

$$\mathbb{E} \left[\frac{|\mathcal{O}_i^{(T)} \cap \mathcal{N}_k^T(i)|}{k} \right] = q_T = 1 - \left(1 - \frac{B-1}{N-1} \right)^T \leq \frac{T(B-1)}{N-1}. \quad (4)$$

Moreover, the realized batch-relational objective depends on the i -th student score row only through $\{c_{ij}^S : j \in \mathcal{O}_i^{(T)}\}$. Scores c_{ij}^S with $j \notin \mathcal{O}_i^{(T)}$ are therefore unidentified by that empirical objective.

Proposition 1 does not imply that every unseen relation must be wrong. It shows that the batch-relational objective cannot determine most teacher-neighbor relations from its realized score arguments; Figure 1 shows that this missing support leads to measurable distortion in practice. The result gives a direct design principle: *teacher geometry should select relational supervision, while the mini-batch should serve only as a computational unit*. A full-corpus objective satisfies this requirement but is impractical to evaluate directly; Section 4 develops a sparse approximation.

4 RIPPLE: CORPUS-ANCHORED GEOMETRY DISTILLATION

RIPPLE implements the requirement derived in Section 3 in four stages: corpus anchoring, multi-scale diffusion, candidate sampling, and cross-anchor row supervision. For a distribution p and a set C with $p(C) > 0$, $p|_C$ denotes restriction to C followed by renormalization.

4.1 CORPUS-LEVEL TEACHER GRAPH

We cache the teacher embeddings and construct a mutual k -nearest-neighbor graph over the corpus. Its support is

$$j \in \mathcal{M}_k^T(i) \iff j \in \mathcal{N}_k^T(i) \text{ and } i \in \mathcal{N}_k^T(j), \quad (5)$$

which retains reciprocal teacher neighborhoods and removes one-directional hub edges. The weighted transition operator is

$$W_{ij}^T = \mathbf{1}[j \in \mathcal{M}_k^T(i)] \exp(c_{ij}^T/\tau_i), \quad P^T = (D^T)^{-1}W^T, \quad (6)$$

where D^T is the row-degree matrix. Unlike a batch-local support, each row of P^T is anchored in the teacher’s corpus geometry. We use an entropy-calibrated bandwidth τ_i to accommodate variation in neighborhood density; its construction is given in Appendix B.2. Once P^T is stored, the teacher is not used during training.

4.2 FROM NEIGHBORHOODS TO MULTI-SCALE TOPOLOGY

The corpus graph corrects the supervision support, but its individual rows describe only immediate neighbors. Global geometry also depends on how these neighborhoods connect. We expose this structure with the lazy transition operator $\tilde{P}^T = \frac{1}{2}(I + P^T)$. For a selected set of scales $\mathcal{R}$ containing 1, the teacher diffusion profiles are

$$g_{i,1}^T = P_i^T, \quad g_{i,r}^T = \delta_i^\top (\tilde{P}^T)^r \Big|_{\mathcal{D} \setminus \{i\}}, \quad r \in \mathcal{R} \setminus \{1\}. \quad (7)$$

Every profile is viewed as a distribution on the common corpus index set, with zero mass at excluded entries. The one-step profile preserves direct teacher neighbors, whereas broader profiles summarize the corpus regions connected to an anchor (Coifman & Lafon, 2006). We use the collection $\{g_{i,r}^T\}_{i,r}$ as a multi-scale representation of teacher geometry.

At each scale, the ideal full-corpus student profile is the direct readout

$$g_{i,r}^S(j) = \frac{\exp(c_{ij}^S/\tau_{i,r})}{\sum_{\ell \neq i} \exp(c_{i\ell}^S/\tau_{i,r})}, \quad j \neq i. \quad (8)$$

This is the ideal full-corpus object approximated by the RIPPLE loss; it does not posit a student Markov walk. We measure the geometry induced by these profiles through

$$d_r^M(i, j) = \text{TV}(g_{i,r}^M, g_{j,r}^M), \quad M \in \{T, S\}. \quad (9)$$

Theorem 1 (Diffusion-geometry preservation). *Fix a scale r . If $\max_i D_{\text{KL}}(g_{i,r}^T \| g_{i,r}^S) \leq \varepsilon_r$, then*

$$\sup_{i,j} |d_r^T(i, j) - d_r^S(i, j)| \leq \sqrt{2\varepsilon_r}. \quad (10)$$

Moreover, for $\bar{\varepsilon}_r = N^{-1} \sum_i D_{\text{KL}}(g_{i,r}^T \| g_{i,r}^S)$,

$$\frac{1}{N^2} \sum_{i,j} |d_r^T(i, j) - d_r^S(i, j)| \leq \sqrt{2\bar{\varepsilon}_r}. \quad (11)$$

Theorem 1 connects the training target to the central goal: aligning anchor profiles controls distortion of the pairwise diffusion geometry. RIPPLE therefore matches teacher diffusion profiles directly at multiple resolutions. The proof is in Appendix A.2; the scale construction and temperatures are deferred to Appendix B.2.

4.3 MASS-AWARE CANDIDATE SAMPLING

Diffusion supplies corpus-level supervision, but it also expands the support of each target. Scoring all N corpus columns per anchor would eliminate the efficiency of mini-batch training. RIPPLE instead samples candidates from the teacher target mass.

Lemma 1 (Mass coverage of target-proportional sampling). *Let p be a distribution and let C_m contain the distinct atoms observed in m independent draws from p . Then*

$$\mathbb{E}[p(C_m^c)] = \sum_j p_j(1 - p_j)^m \leq e^{-mt} + \text{Tail}_p(t), \quad \text{Tail}_p(t) = \sum_{j:p_j < t} p_j, \quad (12)$$

for every $t > 0$.

Lemma 1 shows why this approximation can remain compact: candidate coverage is governed by the concentration of the target, rather than directly by corpus size. RIPPLE applies the result at each scale, retains the highest-mass entries, and samples the remaining mass-aware quota without replacement. Appendix A.3 gives the proof and implementation connection.

For anchor i , the scale-wise draws are merged with hard and random negatives into one shared candidate set C_i . For each scale, define

$$\kappa_{i,r} = D_{\text{KL}}(g_{i,r}^T | C_i \| g_{i,r}^S | C_i), \quad \delta_{i,r}^T = g_{i,r}^T(C_i^c), \quad \delta_{i,r}^S = g_{i,r}^S(C_i^c). \quad (13)$$

Corollary 1 (Sampled control of diffusion geometry). *Let $\eta_{i,r} = \sqrt{\kappa_{i,r}/2} + \delta_{i,r}^T + \delta_{i,r}^S$. For any pair of anchors i, j ,*

$$|d_r^T(i, j) - d_r^S(i, j)| \leq \eta_{i,r} + \eta_{j,r}. \quad (14)$$

Corollary 1 connects the sampled objective back to global geometry. Its bound separates restricted matching error, missed teacher mass, and student leakage. Target-proportional sampling reduces δ^T ; hard and random negatives, together with an ambient comparison term, are used to discourage δ^S . We measure the latter rather than claiming that finite negative samples guarantee full-corpus leakage control.

For an anchor batch B , we pool and deduplicate its candidates,

$$\mathcal{P}_B = \bigcup_{i \in B} C_i. \quad (15)$$

Each text in $\mathcal{P}_B$ is encoded once, while every anchor retains its own target on its sampled support. To expose mass outside the graph-selected candidates, we also use an ambient profile $g_{i,0}^M = \text{softmax}_{j \neq i}(c_{ij}^M / \tau_0)$ for $M \in \{T, S\}$. Let $\mathcal{R}_0 = \{0\} \cup \mathcal{R}$, with $\Omega_{i,0} = \mathcal{P}_B \setminus \{i\}$ and $\Omega_{i,r} = C_i$ for $r \geq 1$. The resulting multi-scale geometry loss is

$$\mathcal{L}_{\text{diff}} = \frac{1}{|B|} \sum_{i \in B} \sum_{r \in \mathcal{R}_0} \omega_r D_{\text{KL}}(g_{i,r}^T | \Omega_{i,r} \| g_{i,r}^S | \Omega_{i,r}). \quad (16)$$

4.4 ROW SUPERVISION FOR CROSS-ANCHOR CONSISTENCY

Mass-aware sampling makes global-profile matching computationally feasible, but a second gap remains. A student embedding induces one symmetric cosine geometry, whereas the softmax loss normalizes every anchor row independently. Perfect row matching can therefore preserve within-row probabilities without fully coupling the underlying similarities across anchors.

Proposition 2 (Softmax gauge and reciprocal-overlap consistency). *Suppose teacher and student one-step rows use the same temperature on a column set Ω_i with $|\Omega_i| \geq 2$. If their row softmaxes coincide, then there is a row-dependent constant a_i such that*

$$c_{ij}^S = c_{ij}^T + a_i, \quad j \in \Omega_i. \quad (17)$$

Construct the reciprocal-overlap graph on supervised rows, with an edge $\{u, v\}$ whenever $v \in \Omega_u$ and $u \in \Omega_v$. The offsets are constant on every connected component of this graph.

Proposition 2 identifies the missing constraint: reciprocal overlap collapses independent row gauges to one component-wise constant. RIPPLE creates these constraints by following the teacher transition graph and supervising the rows encountered along its paths. Starting at an anchor, a walk visits

$$j_0 = i, \quad j_{t+1} \sim P_{j_t}^T. \quad (18)$$

For every visited row u with $|\Omega_u^w| \geq 2$, let $\Omega_u^w = \mathcal{P}_B \cap \mathcal{M}_k^T(u)$ and let $\nu_B(u)$ be its normalized visit count. We add

$$\mathcal{L}_{\text{row}} = \sum_u \nu_B(u) D_{\text{KL}}(P_u^T |_{\Omega_u^w} \parallel \text{softmax}_{v \in \Omega_u^w}(c_{uv}^S / \tau_u)). \quad (19)$$

A realized walk edge supplies a reciprocal constraint whenever both endpoints and cross-columns are present in the supervised pool. Thus the teacher graph not only selects informative columns, but also selects the additional rows needed to connect their normalization gauges. These rows largely reuse existing encodings. Sampling details are given in Appendix B.3, and Proposition 2 is proved in Appendix A.4.

The two components give the RIPPLE objective

$$\mathcal{L}_{\text{RIPPLE}} = \mathcal{L}_{\text{diff}} + \lambda \mathcal{L}_{\text{row}}. \quad (20)$$

The graph, scales, sampling budgets, temperatures, and loss weights are specified in Appendix B. Exact graph construction costs $O(N^2 d_T)$ before sparsification and stores $O(Nk)$ edges; approximate nearest-neighbor search can replace it at larger scale. Each update encodes at most $|B| + |\mathcal{P}_B|$ texts. The teacher graph and all distillation machinery disappear at inference.

5 EXPERIMENTS

We evaluate RIPPLE from two perspectives: whether global geometry alignment improves downstream performance, and whether the gains arise from replacing batch-selected relations with teacher-selected corpus supervision.

Models and data. We evaluate three teacher→student pairs: Qwen3-Embedding-4B → BERT-base (109M), BGE-M3 → MiniLMv2-H768 (66M), and Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M). Following TALAS (Dao et al., 2026), the unlabeled distillation corpus contains approximately 15K sentences sampled evenly from Emotion, WiC, and STS-B. Teacher embeddings are cached once and the teacher is removed before student training.

Evaluation. The nine downstream datasets cover classification (Banking77, Tweet, Emotion), pair classification (MRPC, SciTail, WiC), and semantic textual similarity (SICK, STS12, STS-B). We report macro-F1, average precision, and Spearman correlation, respectively. Emotion, WiC, and STS-B are in-domain; the other six datasets are out-of-domain. “Avg.” is the unweighted mean of the nine primary metrics. Checkpoints and pair-classification thresholds are selected on validation data, and test data are evaluated once.

Baselines and implementation. We compare with the teacher, the undistilled student, SimCSE-unsup, CDM, DSKD, Jasper–Stella, DistillCSE, EMO, and TALAS under the corresponding TALAS protocol (Dao et al., 2026). We use one RIPPLE configuration across downstream tasks and vary only the component named in each ablation. Appendix B gives the complete graph, sampling, optimization, and evaluation settings.

5.1 MAIN RESULTS

Table 1 reports every dataset for all three teacher–student settings. RIPPLE obtains the best mean compact-student score in each setting: 78.09, 77.99, and 74.91. The corresponding margins over the strongest baseline are 0.05, 1.35, and 1.45 points; the first is small relative to run variation. The most consistent gains occur on semantic textual similarity. RIPPLE gives the best compact-student result on STS-B in all three settings and leads all three STS datasets for both MiniLMv2 students. Pair classification is less uniform, including lower WiC scores, so aggregate gains alone do not establish the proposed mechanism.

5.2 WHAT MAKES RIPPLE WORK?

Figure 2 reports downstream Avg., full-corpus teacher–student JS divergence, and held-out eight-step diffusion error along a causal progression. Comparing batch-local KD, random corpus neighbors, and RIPPLE one-step isolates teacher-selected support from the number of comparisons.

Table 1: Results across three teacher $\rightarrow$ student settings on nine datasets. Emotion, WiC, and STS-B (marked with $\star$) are in-domain; the remaining datasets are out-of-domain. Bold and underlined values denote the best and second-best results within each setting, excluding the Teacher and Student base reference rows. TALAS and RIPPLE results are reported as mean $\pm$ standard deviation; the remaining baseline results are from TALAS (Dao et al., 2026). For setting (c), the epoch checkpoint is selected on the validation average.

	Classification (F1)			Pair Classification (AP)			STS (Spearman)			Domain Avg		Avg.
	Banking77	Tweet	Emotion $\star$	MRPC	SciTail	WiC $\star$	SICK	STS12	STS-B $\star$	Avg-In	Avg-Out	Avg.
(a) Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M)												
Teacher	93.63	72.22	68.94	84.79	91.77	66.80	83.43	82.55	86.87	74.20	84.73	81.22
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	42.44	60.33	54.36
SimCSE-unsup	89.21	69.13	53.55	82.86	76.59	71.64	68.19	61.68	70.36	65.18	74.61	71.47
CDM	89.75	70.58	53.11	84.83	75.19	71.06	69.37	68.48	73.92	66.03	76.37	72.92
DSKD	89.66	69.93	54.51	83.13	77.95	<u>71.53</u>	68.65	63.37	71.42	65.82	75.45	72.24
Jasper and Stella	89.71	<u>73.83</u>	65.98	<u>85.33</u>	80.99	70.50	73.96	67.90	75.65	70.71	78.62	75.98
DistillCSE	90.94	70.90	60.80	82.86	78.98	68.63	75.12	72.61	77.08	68.84	78.57	75.32
EMO	90.78	73.54	67.48	84.46	81.57	69.76	75.66	68.83	77.17	71.47	79.14	76.58
TALAS	90.44 $\pm$ 0.13	74.62 $\pm$ 0.17	70.87 $\pm$ 0.52	85.90 $\pm$ 0.10	<u>81.78 $\pm$ 0.29</u>	68.84 $\pm$ 0.15	77.56 $\pm$ 0.10	<u>73.12 $\pm$ 0.30</u>	<u>79.24 $\pm$ 0.10</u>	72.98 $\pm$ 0.19	<u>80.57 $\pm$ 0.01</u>	<u>78.04 $\pm$ 0.07</u>
RIPPLE	91.75 $\pm$ 0.17	73.49 $\pm$ 0.45	<u>70.21 $\pm$ 0.57</u>	83.38 $\pm$ 0.05	83.47 $\pm$ 0.18	66.58 $\pm$ 0.06	<u>77.21 $\pm$ 0.13</u>	75.48 $\pm$ 0.19	81.23 $\pm$ 0.09	<u>72.67 $\pm$ 0.14</u>	80.80 $\pm$ 0.04	78.09 $\pm$ 0.04
(b) BGE-M3 $\rightarrow$ MiniLMv2-H768 (66M)												
Teacher	93.52	73.85	68.56	85.81	91.87	61.47	79.18	78.73	84.87	71.63	83.83	79.76
Student base	81.84	47.70	71.67	79.67	65.06	60.89	49.46	26.76	24.12	44.24	62.41	56.35
SimCSE-unsup	87.89	71.45	55.65	83.87	76.32	67.96	66.93	59.66	63.97	62.53	74.35	70.41
CDM	89.15	70.48	58.71	84.16	75.91	70.00	68.14	62.74	67.78	65.50	75.10	71.90
DSKD	89.51	70.53	57.27	84.39	75.36	<u>69.96</u>	68.81	66.78	70.05	65.76	75.90	72.52
Jasper and Stella	88.38	74.13	62.81	85.98	80.33	68.38	74.35	66.80	74.55	68.58	78.33	75.08
DistillCSE	90.50	72.70	59.88	84.39	78.48	69.49	73.97	70.00	73.73	67.70	78.34	74.79
EMO	90.19	74.26	61.48	85.31	80.82	68.30	75.41	67.06	76.35	68.71	78.84	75.46
TALAS	89.72 $\pm$ 0.19	75.38 $\pm$ 0.11	65.83 $\pm$ 0.58	87.58 $\pm$ 0.02	82.90 $\pm$ 0.12	66.96 $\pm$ 0.07	<u>75.82 $\pm$ 0.11</u>	68.11 $\pm$ 0.20	<u>77.51 $\pm$ 0.18</u>	70.10 $\pm$ 0.13	79.92 $\pm$ 0.05	76.64 $\pm$ 0.04
RIPPLE	90.58 $\pm$ 0.12	73.97 $\pm$ 0.30	71.06 $\pm$ 0.98	85.22 $\pm$ 0.09	83.92 $\pm$ 0.14	62.87 $\pm$ 0.15	78.81 $\pm$ 0.09	73.99 $\pm$ 0.20	81.46 $\pm$ 0.22	71.80 $\pm$ 0.31	81.08 $\pm$ 0.07	77.99 $\pm$ 0.13
(c) Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-H384 (22M)												
Teacher	93.21	71.33	66.53	83.37	90.00	66.85	80.64	77.11	84.60	72.66	82.61	79.29
Student base	68.29	41.07	69.90	78.39	64.56	59.58	47.60	21.95	22.08	40.91	58.45	52.60
SimCSE-unsup	86.67	69.56	54.00	82.29	73.76	65.71	66.04	59.11	62.08	60.60	72.91	68.80
CDM	86.07	70.32	53.04	81.97	72.97	68.34	65.97	61.12	65.75	62.38	73.07	69.51
DSKD	85.66	69.86	52.25	82.17	73.52	<u>68.58</u>	66.54	62.52	66.70	62.51	73.38	69.76
Jasper and Stella	85.94	71.25	57.99	83.36	76.17	68.25	70.57	63.81	70.11	65.45	75.18	71.94
DistillCSE	88.03	69.87	54.24	83.20	77.66	67.88	70.09	68.19	71.53	64.55	76.17	72.30
EMO	87.03	71.67	59.63	83.45	78.17	68.89	70.81	65.78	71.42	66.65	76.15	72.98
TALAS	84.16 $\pm$ 0.15	72.82 $\pm$ 0.11	61.13 $\pm$ 0.52	84.85 $\pm$ 0.10	80.91 $\pm$ 0.27	66.32 $\pm$ 0.07	70.72 $\pm$ 0.15	67.80 $\pm$ 0.08	<u>72.39 $\pm$ 0.31</u>	66.61 $\pm$ 0.24	<u>76.88 $\pm$ 0.07</u>	73.46 $\pm$ 0.10
RIPPLE	<u>87.50 $\pm$ 0.24</u>	70.50 $\pm$ 0.29	66.40 $\pm$ 0.40	81.68 $\pm$ 0.12	<u>80.73 $\pm$ 0.05</u>	65.34 $\pm$ 0.10	73.69 $\pm$ 0.11	72.08 $\pm$ 0.09	76.29 $\pm$ 0.08	69.34 $\pm$ 0.13	77.70 $\pm$ 0.11	74.91 $\pm$ 0.08

Core ablation design.

Six variants: batch-local, random corpus, random graph, RIPPLE one-step, RIPPLE diffusion, and full RIPPLE.

Panels: downstream Avg. $\uparrow$ Full-corpus JS $\downarrow$ Held-out E_8 $\downarrow$.

Figure 2: **Causal ablation of RIPPLE.** Bars show mean $\pm$ standard deviation over seeds. Random corpus support controls for the number of comparisons; a degree-preserving random graph controls for generic graph regularization; the final three variants isolate teacher anchoring, diffusion, and row supervision.

Adding diffusion and then row supervision tests multi-hop and cross-anchor transfer, while a degree-preserving random graph controls for generic graph regularization. All variants share the student, optimizer, schedule, and candidate budget; Appendix C.1 specifies the remaining fairness controls.

5.3 TOPOLOGY COVERAGE AND TRAINING EFFICIENCY

Figure 3 tests the two mechanism–cost trade-offs without adding separate main-text tables. First, varying only k measures how broader teacher anchoring changes retained teacher mass and alignment in fixed teacher-rank bands. Improvement outside the directly supervised band is the relevant multi-hop signal; increasing Mass@ k alone is not. Second, varying only m measures retained target mass, sampled-to-full gradient fidelity, downstream Avg., step time, and memory. Saturation at $m \ll k$ supports the claim that target concentration, rather than corpus size, governs the useful candidate budget. Definitions and complete sweeps are in Appendix C.2.

The appendix provides the controlled settings and exact diagnostic definitions used in both panels.

Mechanism diagnostics.

- (a) k vs. retained teacher mass and fixed rank-band JS.
- (b) m vs. target mass, gradient fidelity, accuracy, time, and memory.

Figure 3: **Coverage and efficiency.** Left: expanding teacher-anchored support should improve alignment beyond the nearest supervised rank band. Right: candidate sampling should retain the full-target signal before its computational cost approaches full-neighborhood scoring.

6 LIMITATIONS AND CONCLUSION

RIPPLE turns a corpus-level geometry target into tractable supervision for a standard text encoder. The coverage analysis explains why batch-local relations under-identify teacher neighborhoods, while the diffusion and sampling results connect the implemented objective back to corpus geometry. Empirically, the strongest and most consistent gains occur on semantic textual similarity; less uniform pair-classification results indicate that geometry preservation does not benefit every task equally.

The method can inherit errors and domain bias from the teacher graph, and its preprocessing is corpus-dependent. Exact neighbor search is quadratic before sparsification, although approximate search can replace it at larger scale. Theorem 1 controls diffusion geometry rather than every property of an embedding space, while Corollary 1 retains an explicit student-leakage term. RIPPLE nevertheless adds no inference-time machinery: deployment uses only the distilled text encoder.

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A PROOFS

Throughout, $\|v\|_1 = \sum_j |v_j|$ and $\text{TV}(p, q) = \frac{1}{2}\|p - q\|_1$. For a distribution p and a set C with $p(C) > 0$, $p|_C$ denotes the conditional distribution on C , extended by zero outside C when two distributions are compared on a common domain.

A.1 PROOF OF PROPOSITION 1

The batch containing anchor i includes $B - 1$ of the remaining $N - 1$ examples uniformly without replacement. Thus, for any fixed $j \neq i$,

$$\Pr[j \in \mathcal{L}_{\mathcal{B}_e}(i)] = \frac{B - 1}{N - 1}. \quad (21)$$

Under independent reshuffling, the probability that j has appeared with i after T epochs is

$$q_T = 1 - \left(1 - \frac{B - 1}{N - 1}\right)^T. \quad (22)$$

Linearity of expectation over the k members of $\mathcal{N}_k^T(i)$ gives Eq. (4); the upper bound follows from $1 - (1 - x)^T \leq Tx$ for $x \in [0, 1]$.

Every realized row- i term in Eq. (2) evaluates student scores only at columns in $\mathcal{L}_{\mathcal{B}_e}(i)$. Across epochs, the objective therefore depends on row i only through $\{c_{ij}^S : j \in \mathcal{O}_i^{(T)}\}$. Any two score arrays that agree on this set have the same realized batch-relational objective, irrespective of their values outside it. This proves the support-level under-identification. $\square$

A.2 PROOF OF THEOREM 1

Fix r and let $a_i = \text{TV}(g_{i,r}^T, g_{i,r}^S)$. The reverse triangle inequality gives

$$|\text{TV}(g_{i,r}^T, g_{j,r}^T) - \text{TV}(g_{i,r}^S, g_{j,r}^S)| \leq a_i + a_j. \quad (23)$$

By Pinsker’s inequality, $a_i \leq \sqrt{D_{\text{KL}}(g_{i,r}^T \| g_{i,r}^S)/2}$. Under the uniform assumption, Eq. (23) is therefore at most $2\sqrt{\varepsilon_r/2} = \sqrt{2\varepsilon_r}$.

For the average statement, Jensen’s inequality yields

$$\frac{1}{N} \sum_i a_i \leq \sqrt{\frac{1}{2N} \sum_i D_{\text{KL}}(g_{i,r}^T \| g_{i,r}^S)} = \sqrt{\frac{\varepsilon_r}{2}}. \quad (24)$$

Averaging Eq. (23) over ordered pairs (i, j) gives $2N^{-1} \sum_i a_i \leq \sqrt{2\varepsilon_r}$, completing the proof. $\square$

A.3 PROOFS FOR SAMPLED SUPERVISION

Proof of Lemma 1. An atom j is absent from m independent draws from p with probability $(1 - p_j)^m$. Hence

$$\mathbb{E}[p(C_m^c)] = \sum_j p_j (1 - p_j)^m. \quad (25)$$

For any $t > 0$, atoms with $p_j \geq t$ contribute at most $e^{-mt} \sum_{j:p_j \geq t} p_j \leq e^{-mt}$; the remaining atoms contribute at most $\text{Tail}_p(t)$. This proves Eq. (12). $\square$

The same upper bound applies to target-proportional sampling without replacement: conditional on not having selected j , its next-draw probability is at least p_j . RIPPLE additionally includes deterministic high-mass candidates and negatives, so we measure the realized missed target mass $\delta_{i,r}^T$, rather than treating the bound as an equality for the full sampler.

Lemma 2 (Restriction to a candidate set). *Let p and q be distributions on a common finite set. If $p(C^c) = \delta_p$ and $q(C^c) = \delta_q$, then*

$$\text{TV}(p, q) \leq \text{TV}(p|_C, q|_C) + \delta_p + \delta_q. \quad (26)$$

Proof. Extending the conditional distributions by zero outside C gives $\text{TV}(p, p|_C) = \delta_p$ and $\text{TV}(q, q|_C) = \delta_q$. The claim follows from the triangle inequality. $\square$

Proof of Corollary 1. Pinsker’s inequality and Lemma 2 imply

$$\text{TV}(g_{i,r}^T, g_{i,r}^S) \leq \sqrt{\kappa_{i,r}/2} + \delta_{i,r}^T + \delta_{i,r}^S = \eta_{i,r}. \quad (27)$$

Applying Eq. (23) to anchors i and j gives Eq. (14). $\square$

A.4 PROOF OF PROPOSITION 2

Let the coincident teacher and student softmax rows on Ω_i be p_i . For any $j, j' \in \Omega_i$, equality of log-ratios gives

$$\frac{c_{ij}^S - c_{ij'}^S}{\tau_i} = \log \frac{p_i(j)}{p_i(j')} = \frac{c_{ij}^T - c_{ij'}^T}{\tau_i}. \quad (28)$$

Thus $c_{ij}^S - c_{ij'}^S$ is constant in j on Ω_i ; denote it by a_i . Conversely, adding a constant to every logit in a row leaves its softmax unchanged.

For an edge $\{u, v\}$ of the reciprocal-overlap graph, Eq. (17) gives $c_{uv}^S = c_{uv}^T + a_u$ and $c_{vu}^S = c_{vu}^T + a_v$. Teacher and student cosine scores are symmetric, so $a_u = a_v$. Equality propagates along paths, making the offset constant on each connected component. $\square$

Component	Configuration
Teacher graph	Mutual k NN, $k = 200$; target perplexity $\rho = 30$
Diffusion	Scales $\mathcal{R} = \{1, 2, 4\}$; truncation tolerance 0.01
Candidates	66 per anchor: 14 diffusion, 26 hard, 26 random
Optimization	AdamW; batch size 32; 5 epochs; learning rate 2×10^{-5}
Schedule	6% warm-up; cosine decay to 3×10^{-6}
Row supervision	4 walks of length 4; starts after epoch 1; $\lambda = 0.5$
Ambient profile	Temperature $\tau_0 = 0.10$

Table 2: Canonical RIPPLE configuration. A controlled experiment changes only the component named in that experiment.

B IMPLEMENTATION AND TRAINING DETAILS

We report the core configuration used for RIPPLE and the implementation choices deferred from Section 4. Unless an ablation explicitly varies one factor, the same configuration is used across all downstream tasks and teacher–student pairs.

B.1 TRAINING CONFIGURATION

Teacher and student profiles use the same temperature at each corresponding scale. The teacher embeddings and graph targets are computed once before training; the teacher is not loaded during student optimization.

B.2 GRAPH AND DIFFUSION CONSTRUCTION

Entropy-calibrated bandwidth. For a mutual-neighborhood row, let $p_j(\beta) \propto \exp(\beta c_{ij}^T)$, where $\beta = 1/\tau_i$. We choose τ_i such that

$$H(P_i^T) = \log \rho. \quad (29)$$

Because $dH_i/d\beta = -\beta \text{Var}_{p(\beta)}[c_{ij}^T]$, the entropy is monotone and the non-degenerate solution is found by bisection. If row degree $d_i \leq \rho$, we clip the target entropy to $(1 - 10^{-3}) \log d_i$; degree-one and constant-score rows use the uniform distribution on their available support.

Scales and sparse profiles. We set $\tau_1(i) = \tau_i$ and $\tau_r(i) = \sqrt{r} \tau_i$ for broader scales, with loss weights $\omega_r = 1/r$ and $\omega_0 = \omega_1$. After each sparse diffusion step, we retain the smallest support carrying at least $1 - \delta$ mass, with $\delta = 0.01$, and renormalize. This rule controls the stored profile width without introducing an additional scale-specific budget.

B.3 CANDIDATE AND ROW SAMPLING

The diffusion quota is split as evenly as possible across scales, with the remainder assigned to sharper scales. At each scale, RIPPLE retains the two highest-mass entries and draws the remaining quota without replacement in proportion to teacher mass. Hard negatives are the nearest teacher examples outside the diffusion pool, prioritizing the same source corpus; the remaining negatives are sampled uniformly. Candidates are deduplicated, pooled across anchors, and encoded once as in Eq. (15).

Non-backtracking walks. At step $t + 1$, the walk samples from $P_{j_t}^T$ conditioned on $j_{t+1} \neq j_{t-1}$. The exception is a degree-one node, where the previous edge is the only available transition. Avoiding immediate returns increases the number of distinct supervised rows at a fixed walk budget. A walk selects rows only: for every selected row, Eq. (19) uses its available dense teacher target rather than a single sampled transition, reducing gradient variance without changing the expected row objective.

C CONTROLLED EXPERIMENT DETAILS

C.1 CORE ABLATIONS

All variants in Figure 2 use the same student initialization, optimizer, epochs, seeds, and number of scored anchor-column pairs. *Batch-local KD* normalizes teacher and student similarities over co-occurring examples. *Random corpus neighbors* uses a uniformly sampled corpus support with the same size as the one-step RIPPLE support. *RIPPLE one-step* uses only $r = 1$ and no row loss; *RIPPLE diffusion* adds $r \in \{2, 4\}$; and *Full RIPPLE* additionally enables $\mathcal{L}_{\text{row}}$.

The random-graph control applies degree-preserving double-edge swaps to the undirected teacher graph. Edge weights are recomputed from teacher similarities at the rewired endpoints, and row bandwidths are recalibrated to the same perplexity ρ . It then uses the same scales, sampler, and row loss as Full RIPPLE. This preserves graph size, degree sequence, and optimization while removing teacher-neighbor adjacency.

For a fixed diagnostic anchor set $\mathcal{A}$, held-out diffusion error is

$$E_r = \frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} \text{TV}(g_{i,r}^T, g_{i,r}^S), \quad r \in \{1, 8, 16\}. \quad (30)$$

Scales 8 and 16 are evaluated but never supervised. Full-corpus JS and student leakage use the same anchors and corpus columns for every variant.

C.2 COVERAGE AND SAMPLING DIAGNOSTICS

Neighborhood breadth. We vary only $k \in \{50, 100, 200, 400\}$ and hold the candidate budget, scales, row weight, optimizer, and schedule fixed. Retained ambient teacher mass is

$$\text{Mass@}k = \frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} g_{i,0}^T(\mathcal{M}_k^T(i)). \quad (31)$$

We also measure teacher-student JS divergence in the fixed teacher-rank bands 1:50, 51:200, and 201:800, and over the full corpus. The bands do not change with k .

Candidate budget. We vary only the number m of candidates. On a fixed checkpoint and probe batch, sampled-to-full gradient fidelity is

$$G_m = \cos\left(\nabla_{\theta} \mathcal{L}_{\text{diff}}^{(m)}, \nabla_{\theta} \mathcal{L}_{\text{diff}}^{(\text{full})}\right). \quad (32)$$

Target mass is measured before renormalization. Step time excludes graph preprocessing and is averaged after warm-up; peak accelerator memory is reset immediately before the measured steps. All variants use the same hardware and software stack.